

AI TOOLS USAGE AND PERCEIVED LEVEL OF CRITICAL THINKING SKILLS OF SENIOR HIGH SCHOOL STEM STUDENTS



INTRODUCTION

The study investigated how the use of Artificial Intelligence (AI) tools affects the critical thinking skills of Senior High School STEM students in Davao City. Researchers explored whether AI improves learning efficiency and problem-solving or causes overdependence that weakens creativity and independent thinking. The study focused on five AI usage factors: utility, expectancy, attainment, intrinsic value, and cost, and six critical thinking skills: remembering, understanding, applying, analyzing, evaluating, and creating.

LITERATURE REVIEW

Existing studies show that AI tools like chatbots and adaptive learning systems help students by:

- Personalizing learning experiences
- Improving research and problem-solving
- Enhancing efficiency and engagement
- Supporting higher-order thinking skills

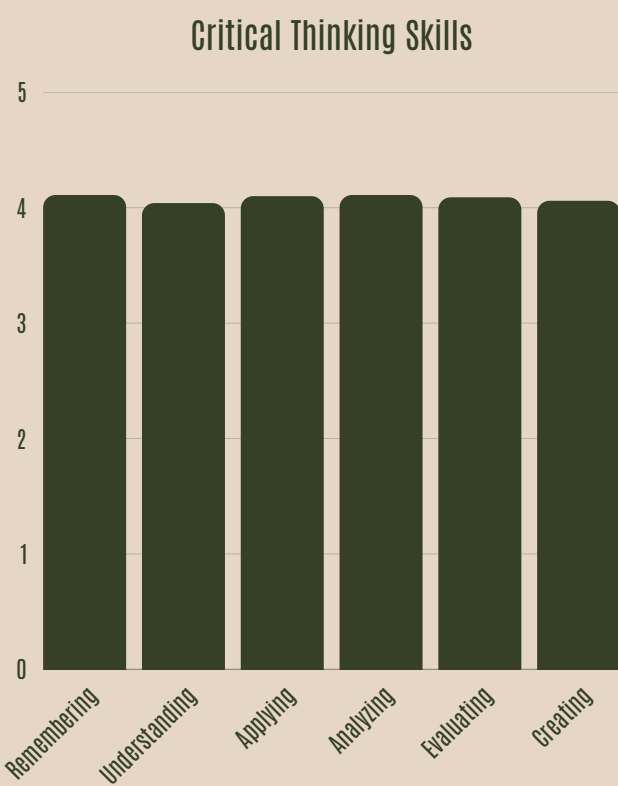
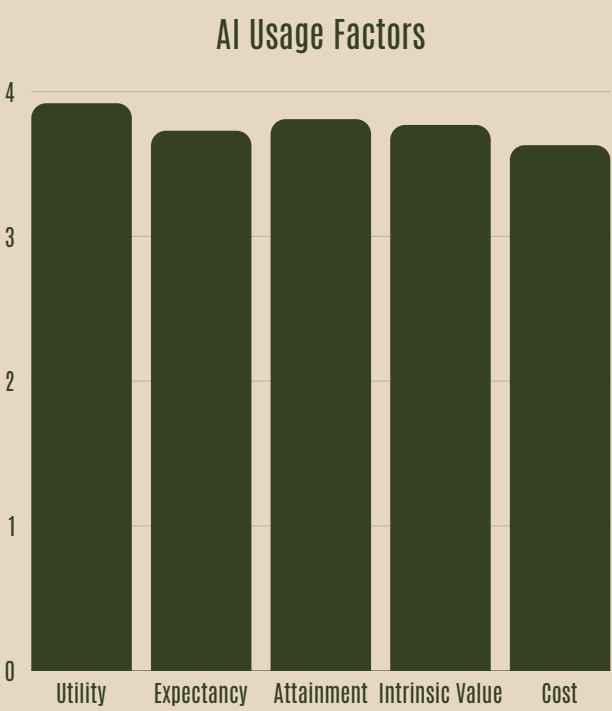
However, literature also warns that excessive AI reliance may reduce creativity, independent analysis, and authentic learning. Researchers emphasized balancing AI integration with traditional critical thinking activities and ethical AI use. The study used Self-Determination Theory and Bloom’s Revised Taxonomy as theoretical foundations.

METHODOLOGY

- Research Design: Quantitative descriptive-correlational study
- Respondents: 298 Grade 11 and 12 STEM students from a private school in Davao City
- Sampling Method: Random sampling using Slovin’s Formula
- Data Collection:
 - a. Distribution of online survey questionnaires
 - b. Collection of consent from participants
 - c. Evaluation of AI usage and critical thinking skills using Likert-scale surveys
- Data Analysis Tools:
 - Mean
 - Standard Deviation
 - Spearman Rank Correlation

The study measured the relationship between AI tool usage and perceived critical thinking skills.

RESULTS



- Statistical analysis showed a significant relationship between AI usage and critical thinking skills:
- p-value = 0.0000000000000001115 (< 0.05)

CONCLUSION

The study concluded that AI tools positively influence the critical thinking skills of STEM students when used responsibly. AI improves convenience, engagement, analysis, evaluation, and problem-solving abilities. However, overdependence on AI may weaken students’ originality and ability to generate their own ideas. Researchers recommended integrating AI ethically into education through teacher training, responsible student use, and balanced learning activities that still encourage independent critical thinking.